

PART 1: AI CONCEPTS

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence. These tasks include learning, problem-solving, decision-making, understanding language, recognizing images, and generating content.

AI systems work by analyzing data, identifying patterns, and making predictions or decisions based on the information they receive. Today, AI is used in many applications such as chatbots, recommendation systems, self-driving cars, voice assistants, medical diagnosis, and content generation tools.

The main goal of AI is to make machines smarter and more capable of assisting humans in different fields.

Image — AI Robot



Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Technology	Description	Example
Artificial Intelligence (AI)	A broad field where machines perform intelligent tasks similar to humans.	Voice assistants
Machine Learning (ML)	A subset of AI where systems learn from data without explicit programming.	YouTube recommendations

Technology	Description	Example
Generative AI	AI that creates new content such as text, images, music, or videos.	ChatGPT
Agentic AI	Advanced AI systems that can make decisions and perform tasks independently.	AI agents

AI is the main concept, while ML, Generative AI, and Agentic AI are specialized branches inside AI.

Q3. Explain any three open-source AI models available on Hugging Face

1. Llama 3

Meta developed Llama 3 as an open-source large language model. It can answer questions, generate content, summarize text, and help developers build AI applications.

2. Stable Diffusion

Stability AI created Stable Diffusion for generating images from text prompts. It is widely used for digital art, posters, and AI image generation.

3. Mistral

Mistral AI developed Mistral models for efficient and fast text generation tasks. These models are lightweight and useful for developers with limited hardware resources.

These models are available on [Hugging Face](https://huggingface.co) where developers can access and use AI models freely.

Image — Artificial Intelligence Technology



Q4. What is the difference between traditional programming and AI-assisted development?

Traditional Programming

Developers manually write all logic and rules.

Output depends on predefined instructions.

More manual effort is required.

Mostly rule-based systems.

AI-Assisted Development

AI tools help generate code automatically.

AI learns patterns from data.

Faster and more efficient development.

Intelligent and adaptive systems.

Traditional programming follows exact instructions written by programmers, while AI-assisted development uses intelligent tools to improve coding, automation, and productivity.

Q5. Explain AI hallucinations with one example

AI hallucination happens when an AI system generates false or incorrect information but presents it as if it were true. This usually happens when the AI lacks proper information or misunderstands the topic.

For example, if a chatbot is asked about a book that does not exist, it may still create a fake author name, publication date, and summary instead of admitting that the book is unavailable.

AI hallucinations can spread misinformation, so users should always verify important AI-generated information.